STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 7: Gravitation

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculations and units. Original practice material, not an official examination.

Section A - 1-mark questions - 8 x 1 = 8 marks

Kepler’s law of equal areas? [1]
Universal gravitation law? [1]
SI unit of G? [1]
g at surface of a spherical planet? [1]
Gravitational potential energy zero at infinity? [1]
Does escape speed depend on launched mass? [1]
What supplies centripetal force to a satellite? [1]
Kepler’s third law for fixed central mass? [1]

Section B - 2-mark questions - 6 x 2 = 12 marks

Force between 2 kg and 3 kg 1 m apart; G=6.67x10^-11. [2]
Weight of 50 kg for g=9.8? [2]
If orbit radius doubles, circular-speed factor? [2]
At altitude h=R, ratio g_h/g? [2]
Approximate potential-energy change for small height h? [2]
If orbital radius becomes 4r, period factor? [2]

Section C - 3-mark questions - 4 x 3 = 12 marks

Find g at height h=R/2 relative to surface g. [3]
Escape speed for g=5 m/s^2 and R=2x10^6 m? [3]
GM=4x10^14, r=7x10^6 m: orbital speed? [3]
Weight is 600 N at surface. Weight at h=R? [3]

Section D - 4-mark questions - 2 x 4 = 8 marks

Derive circular orbital speed and period. [4]
1000 kg satellite at r=7x10^6 m, GM=4x10^14. Find speed and energy. [4]

End of paper - check working, units, signs and significant figures.